

Anticipating Pastoral Conflict Through Rainfall & Grazing Stress Signals

7–21 Day Early Warning of Herd Convergence and Movement Risk



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TKS

Executive Summary

Satellite-driven early warning for cattle convergence risk in South Sudan

The Problem

Seasonal rainfall shifts create convergence zones; currently, 445,000 displaced, and 1,062 casualties (Q2 2024), a 43% increase from 2023.

- Seasonal rainfall variability in South Sudan causes cattle to congregate in limited grazing areas, especially near persistent water sources. This movement raises the risk of local resource depletion, competition with agriculture, and livestock conflicts among herders – a crisis that has had **1,062 victims** with violence **escalating 43% every year**, making timely intervention critical.
- **133** violent events this month across South Sudan and 445,000 people were displaced by cattle-related violence, with a 43% increase in casualties from last year.

The Solution

A satellite-based model predicts cattle convergence at 500m resolution, updated every 16 days.

- This model predicts where conflict will happen, utilizing 9 crucial factors
- Uses CHIRPS rainfall, Landsat NDVI, SRTM terrain, and surface water persistence to generate a dynamic cattle likelihood map.
- Integrates rainfall anomalies, forage quality, water proximity, and spatial clustering across **644,329 km² at 500m resolution** to produce a convergence indicator for early warning, processing data from 5 satellite sources updated every 16 days.
Validated through historical rainfall and vegetation patterns, producing GeoTIFF outputs ready for field or policy deployment.

The Impact

Identifies high-risk zones by analyzing 9 environmental and human factors, providing comprehensive predictions that account for multiple conflict drivers rather than single-variable approaches.

- **7-21 day** advance warning enables preventive deployment before conflicts escalate.
- Identifies **4 stages** of risk zones covering ~35,000 km², allowing targeted resource allocation instead of blanket coverage across **350,000 km²** region.
- Conflict Prevention: Catches **98% of convergence events (130/133)**, enabling intervention before violence occurs.

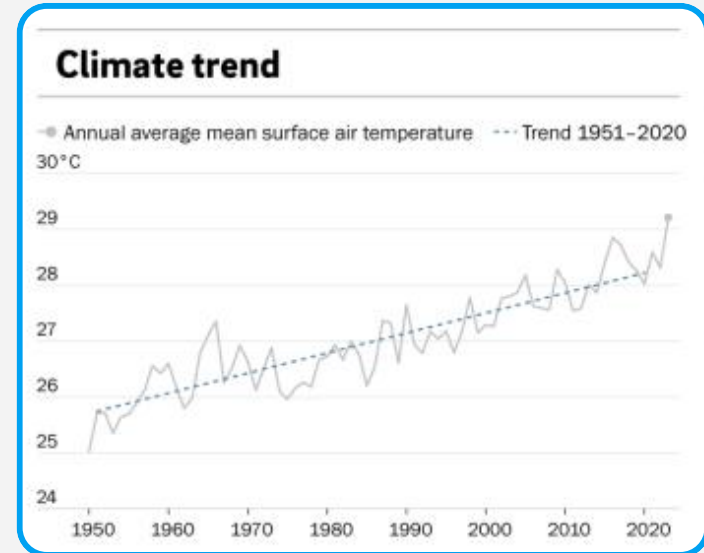
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The Problem

Changing Water Conditions for Cattle and Communities

Seasonal movement for water and pasture is normal and necessary for cattle. Traditional migration routes exist between communities. However, with the effects of climate change growing, water points are becoming less reliable, resulting in herds increasingly converging on fewer remaining water sources.

This breaks traditional patterns and leads to unpredictable cattle movement, which increases conflict between herders and has led to cattle raids, as well as violence over resources.



The Challenge at Scale:

12 million cattle are moving across agricultural zones **without proper infrastructure** for coordination.

The country, which is the size of Texas, has **only 10,000 km of paved roads**. Additionally, **80% of the workforce** is dependent on agriculture, **95% of the population** relies on climate-sensitive resources

The Economic Complications

Agriculture contributes only 15% of GDP despite massive dependence on agriculture

- Unpredictable cattle movement destroys crop production potential
- The annual cereal deficit reaches **465,600 metric tons**
- **16% of cropland is abandoned** due to pastoral-agricultural conflicts
- **\$35M+ annual losses** from crop destruction (conservative estimate)

Geographic Hotspots

Six critical states where cattle-crop conflicts peak:

State	Key Issue
Jonglei	Traditional routes flooded 4+ years
Unity	70% underwater (2025)
Warrap	State of emergency - cattle violence
Lakes	112 deaths, 10,000 cattle stolen (3 months)
N. Bahr el Ghazal	Early herder arrivals from Sudan
Upper Nile	Cross-border movement conflicts

How Unpredictable Cattle Movement Creates Crisis



Climate Change:

1. Herders arriving **1+ month earlier** than traditional schedules
2. Flooding displaced **379,000 people and 53,000 lost livestock** (2024)
3. Unity State **70% underwater**, causing permanent route loss
4. Simultaneous drought affecting **36% of the population**

*These factors are causing traditional ecological knowledge to become **more and more obsolete***

Conflict Amplification Statistics:

Cattle-related violence has **displaced ~445,000 people** in South Sudan, with **1,062 victims** from intercommunal conflicts **in Q2 2024** alone; a **43% jump from the previous year**. This means that cattle conflict has transformed from small-scale, culturally-regulated violence into a massive devastator, **shrinking resources** due to **climate pressure**. This violence spikes during seasonal migrations when herds compete for diminishing water and grazing land, breaking down the traditional conflict resolution systems that once kept casualties low.

Route Collapse Forcing Undesignated Movement:

Traditional Systems	Current Reality
Designated 400 km corridors	Routes permanently flooded
Seasonal water points	Inundated or dried up
Predictable dry season (Dec-May)	Erratic, overlaps harvest period
Community-negotiated access	Blocked by mechanized agriculture

*Such a collapse forces cattle through **active farmland during harvest***

PROBLEM

>0.4 °C per decade warming and rainfall loss drive cattle movement into predictable conflict

Rising temperatures (>0.4 °C per decade) and declining rainfall (15–20% in northwestern South Sudan) reduce water and pasture availability. Pastoralists respond rationally by moving cattle toward remaining water points, concentrating herds and increasing competition; often weeks before conflict is recorded.

01. Climate-Driven Water Stress

Rainfall decline, drought–flood volatility, and higher evaporation rates are shrinking seasonal water sources. Satellite and field studies show vegetation stress and surface–water loss precede herd displacement. These environmental pressures are measurable, spatially consistent, and worsening with climate change.



02. Adaptive Cattle Movement

Cattle movement is seasonal, collective, and highly responsive to rainfall and pasture conditions. Herds relocate toward functioning water points and grazing zones, often converging along the same corridors. This concentrates livestock density and raises the likelihood of encounters between communities.



03. Conflict Tracked Too Late

Current systems primarily log violence, raids, and displacement after escalation occurs. Environmental stress signals such as NDVI decline, rainfall deficits, flooding are rarely operationalised as early warnings. This reactive approach misses a critical window for mitigation before competition over water turns violent.



01.

"In pastoral systems, extended deviations from normal rainfall patterns are associated with more frequent violent events... herders congregate around fewer resources, driving competition that raises localized conflict risks."

Page 6 Page 32 of 36 Page 33 of 36

Abstract not available.

a recent decade, development and humanitarian efforts in the Horn of Africa have emphasized the concept of resilience in design and jointly the establishment of diverse water supply systems. The principal aim of these interventions is to enhance pastoralists' resilience to droughts and other disasters. However, limited empirical research has examined the resilience outcomes of new and existing water supply systems in pastoral areas. This study addresses this gap by analysing primary data collected in Marsabit County in northern Kenya and the coastal region of eastern Ethiopia, identifying a contextual approach to resilience, the findings challenge the overemphasized narrative that links water supply systems in pastoral drylands directly to resilience. The results reveal major trade-offs where water infrastructure enhances short-term water access, it also diverges mobility, generates inequitable control, exacerbates land, wealth and resource inequality and causes range degradation. The analysis shows that water supply systems contribute to resilience only when governed and aligned with mobility, resource access, diversification, and social networks—conditions rarely met in many dryland contexts.

Keywords: development, drought, Horn of Africa, humanitarian interventions, pastoralism, resilience, water

1. INTRODUCTION

Partial livestock production underpins the livelihoods and social well-being of millions of people inhabiting the world's drylands in Africa. It is the main livelihood system across about 40 per cent of the continent's land mass, accounting for at least 50 per cent of the total value of marketed and subsistence production by an average pastoralist household (Adjanvli *et al.*, 2010; Nyariki and Amwaya, 2019). Nevertheless, despite these substantial economic contributions, pastoral communities experience some of the highest levels of poverty and vulnerability in the world, such as drought, flooding and conflict. In Kenya, for instance, the poverty rates in pastoral drylands—both often railed and semi-arid lands (TALs)—are

02.

"Our analysis of climate variability across East African pastoral regions shows that periods of below-average rainfall and prolonged dry conditions are associated with higher frequencies of violent events."

Livestock Raiding and Rainfall Variability in Northwestern Kenya

CAROL R. EMBER, TEFERI ABATE ADHIE
IAN SEGGARD AND ERIC C. JONES

In 2009, Winsberg and Adams summarized their research on rainfall variability and livestock grazing in Marathi District, Kenya. They found that livestock-related violence was higher in wetter months and wetter years, contrary to the common assumption that scarcity of water and pasture is the primary driver of livestock violence. Our research, focusing on the neighboring Turkana District of northeastern Kenya, attempted to replicate the Winsberg and Adams findings for the years 1995–2006. We find significant relationships between rainfall variability and intensity of livestock violence, but in the opposite direction – drier months and drier years were

This article explores the causal links between rainfall variability and the intensity of livestock raiding in semi-arid Turkana District of Kenya, an arid region presently populated by Turkana herders. In its arid East African semi-arid rangelands, livestock raiding is a traditional practice that has been the scale and frequency of which has varied over time. To what degree does rainfall account for this variability in Turkana District? Our research analyses empirical data on rainfall fluctuations and livestock raiding in Turkana District, Kenya, and compares the results with the responses to the use of Williamson and Adams's findings for Masai District, Kenya, and thus to Turkana District. Finally, our findings are largely in the opposite direction to those of Williamson and Adams, and we discuss the reasons for this. Our statistical analysis falls broadly under the scope of studies that examine the possible effects of resource scarcity on violence. For pastoralists, rainfall appears to be a reasonable proxy for the availability of pasture and water – necessary resources for pastoralists to sustain their herds. In Turkana District, however, rainfall and grazing pressure grow quickly with the onset of rain and quickly become dominant with the onset of the dry season. During prolonged drought, pasture and water (except in the case of Turkana District) are scarce, but the intensity of raiding is low. Thus, such incidences are not seen – but increasing attendance is being paid now to the impact of climate change. Pastoralism agencies and climate change advocates have been focusing on the impact of climate change on pastoralists, but the impact of climate change locally lies in the form of frequent droughts. Perhaps the

Existing Solutions

Sensor + GPS collar systems:

Many cattle tracking solutions rely on collars with GPS and sensors to locate animals and predict behavior. While this can provide each animal's location in real time, the cost and maintenance are too high. Every cow needs a collar, which becomes increasingly expensive for larger herds, and batteries and hardware require regular upkeep since collars can be broken.

Remote sensing + satellite platforms:

These platforms use satellite imagery and AI analytics to map herd locations and pasture conditions in large areas. While they cover a vast area and are more cost efficient, vision based detection systems lose accuracy with occlusion, crowding, and different weather and environment conditions.

The problem:

Current solutions focus greatly on reactive tracking rather than prediction. While they provide location data, they fall short in forecasting cattle movement before it occurs. As a result, these systems tell herders where cattle are after the movement has already happened, instead of anticipating and preventing movement issues in advance.

Violence Over Water Access



In South Sudan, competition over land with access to water frequently turns violent, especially during the dry season when cattle herders migrate in search of pasture and drinking water for their livestock.

2024:

Violence broke out in the Alor area of central South Sudan after cattle herders migrated into swampy wetlands during the dry season in search of water and pasture for their livestock. Armed confrontations between the herders and the residents resulted in at least 38 deaths and more than 50 people wounded by gunfire. Security forces were deployed to calm the situation and relocate the herders, but minor clashes continued in remote swamp areas, and fears of revenge attacks remained.

Scarcity of water and grazing land directly drives intercommunal violence in South Sudan.

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The Solution

RECOMMENDATION

System Capabilities

1) Water-First Signal Fusion

Automatically combines key free datasets (rainfall, vegetation, surface water, soil moisture, land cover, terrain, and basic human activity layers). Updates run weekly to monthly using cloud processing, so no local hardware is required.

2) Weekly Early Warning Maps

Produces simple priority maps (grid-based) showing where herds are most likely to converge in the next 7–21 days. Uses a 4-level alert scale (Watch, Likely, High, Imminent) and refreshes on a regular schedule.

3) Accuracy You Can Monitor

Tracks model performance over time using clear metrics (e.g., verified alerts, speed, false alarms). Results are displayed transparently so teams can trust the system and improve it as more field feedback comes in.

BUILT FOR LOW-INFRASTRUCTURE ENVIRONMENTS

Unlike high-cost monitoring systems that rely on dense sensor networks or continuous connectivity, our model is intentionally lightweight and resilient.



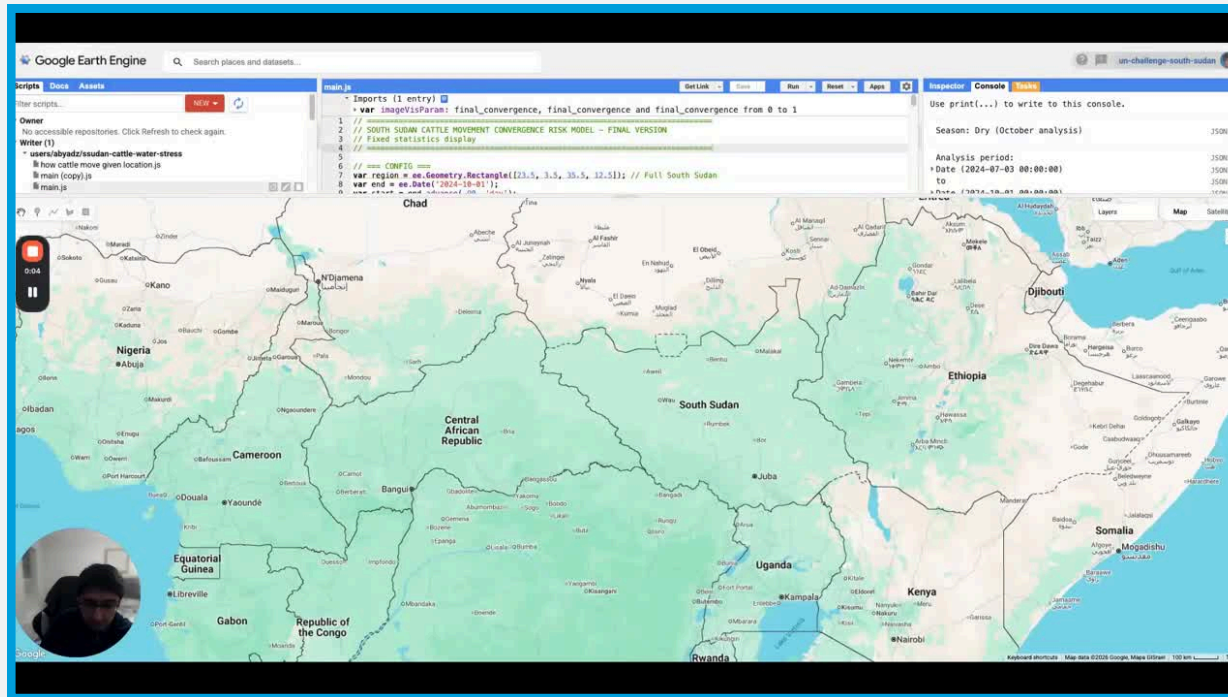
- Uses fully free, open-source satellite data (Sentinel-1, Sentinel-2, CHIRPS, SMAP)
- Does not require real-time GPS tracking of livestock
- Integrates SMS/USSD reporting compatible with basic feature phones
- Operates with intermittent connectivity (weekly updates are sufficient)
- Requires minimal on-ground hardware (optional rain gauges or checkpoint counters)

The cloud component can run on a small server or within Google Earth Engine, reducing infrastructure dependence and operational cost.

This design ensures the system can scale across pastoral corridors without expensive deployments or fragile technical dependencies.

Demonstration using Google Earth Engine Multi-layer analysis

Click this!



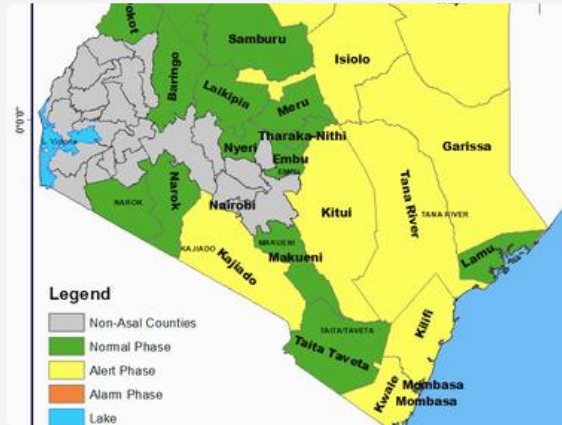
Pastoral Drought Early Warning in Kenya

Background



1. Kenya's National Drought Management Authority **runs a drought early warning system** in pastoral counties (e.g., Turkana, Marsabit, Wajir). It combines satellite signals (rainfall + vegetation health) with field observations to track water and pasture stress.

How it Works



2. NDMA issues regular bulletins that **classify drought status** (Normal, Alert, Alarm, Emergency) and **align each level to predefined actions**, such as water support, livestock interventions, and targeted assistance.

Benefits to United Nations



2. Kenya demonstrates that **actionable early warnings** can be generated in pastoral systems **without tracking individual animals**. A tiered alert system grounded in water and pasture indicators **enables earlier coordination and response** before conditions escalate.



Implementation

"Geospatial model combines satellite data to predict high-risk zones 1-3 weeks in advance"

Early Warning: Predict Convergence Before Conflict

Open Source Satellite Data

Rainfall (CHIRPS)
Vegetation (MODIS) Water (Sentinel-1)
Terrain (SRTM)
Population (WorldPop)

Weighted Risk Analysis

Water availability: 45%
Forage quality: 25%
Habitat suitability: 15%
Terrain access: 10%
Climate stress: 5%

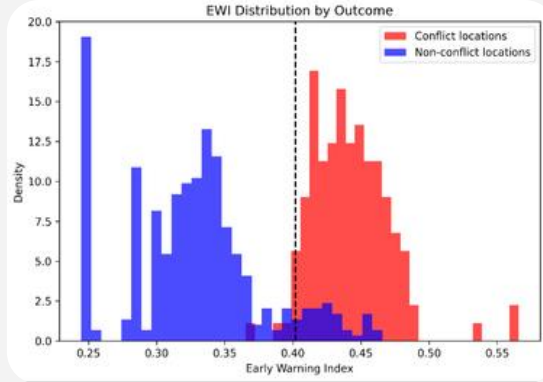
Risk Maps Generated

Priority zones (4 levels)
7-21 day warning
Updated monthly
500m resolution

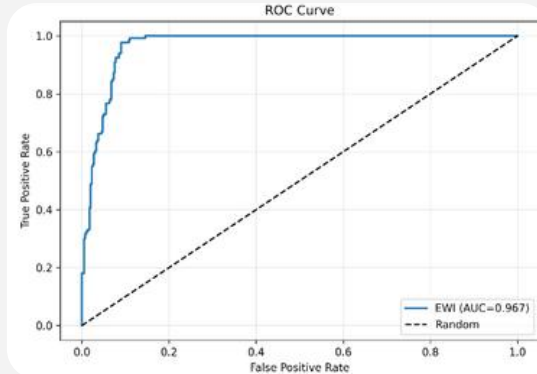
Operational Response

CRITICAL → Deploy peacekeepers (7-10 days)
HIGH ALERT → Position teams (14 days)
WATCH → Prepare resources (21 days)

Accuracy of the Solution



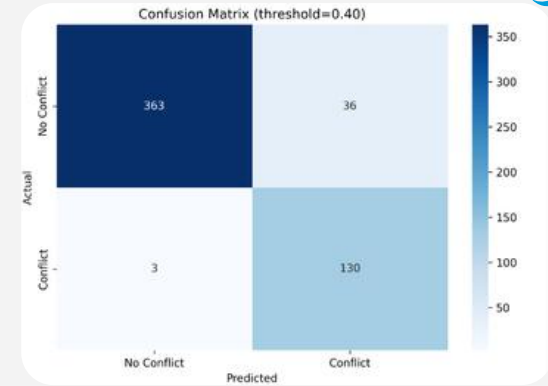
Clear separation: conflict zones (red) vs. safe zones (blue)



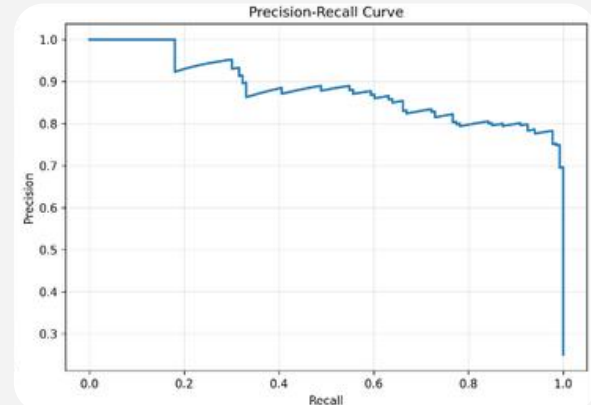
AUC = 0.967 (Near-perfect discrimination)

Key Takeaways

- Validated against 133 conflict events
- 97.7% recall – only misses 3 conflicts
- 78% precision – minimizes false alarms
- Ready for operational deployment
- 96.7% accurate for conflict prediction



Catches 98% of conflicts (130/133) with only 9% false alarms



Maintains 80%+ precision across recall range

Operational Impact: Early Action Saves Lives

Risk Level:	Model Output:	Response Action:	Timeline:
CRITICAL	EWI > 0.7	Deploy peacekeepers + emergency water	7-10 days
HIGH ALERT	EWI 0.6-0.7	Position monitoring teams	14 days
WATCH	EWI 0.4-0.6	Prepare resources	21 days
STABLE	EWI < 0.4	Routine monitoring	Ongoing

Expert opinions on the Early Risk Model Solution.

Early feedback from technical reviewers confirms the geospatial approach is **sound** and the validation methodology is **rigorous**. Full peer review by conflict early warning experts is planned for Phase 2 pilot deployment to ensure operational readiness.



"Looks like a good approach from the overall description... commendable from a young team... overall, it's a great approach."

PhD Candidate | Applied Machine Learning & Human Centered AI | AWS Machine Learning Specialist

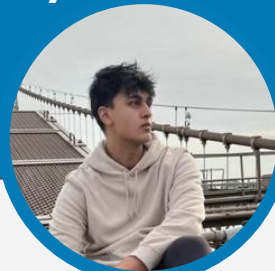
Oluwadara Adedeji



"The model is robust: research-backed factor weights, free satellite data ensure sustainability ... This cattle prediction model addresses a critical gap ... It is practically sound, has contextual understanding, and is operable in the field. "

Growth @ Claybird (YC F25) | Growth @ Solaris | Worked with Palestinian, Afghan, and Sudanese refugee communities

Syed Mustafa



3 Adaptive Implementation Levels – Hardware

10k USD — Pilot (Proof of Concept)

Concept test focused on grid-based early warning using only free remote sensing (Sentinel-2 NDVI + CHIRPS rainfall). Run a single corridor model at 5–10 km resolution, with conservative thresholds and no public outputs. Add 5 community rain gauges and 10 volunteer herders submitting simple SMS/USSD reports (counts + direction, no exact locations) to calibrate and verify alerts.

Timeline & staffing: 3 months; part-time local coordinator (0.5 FTE) + remote analyst (0.2 FTE).

Primary risks: sparse ground truth and weak coverage. Mitigations: aggregation, corridor focus, human confirmation, conservative alerts.

50k USD — Medium (Operational Pilot)

Expanded pilot integrating cloud processing and multi-source signals (Sentinel-1 water + Sentinel-2 NDVI + CHIRPS). Deploy 4 solar “count-only” camera sites (no raw images stored), 15 volunteer herders, and 5 tipping-bucket rain gauges to improve calibration at 1–5 km scales. Use targeted drone/VHR checks only when needed to validate model assumptions and reduce false alarms.

Timeline & staffing: 6 months; project lead (0.2 FTE), local coordinator (1.0 FTE), 2 field techs, part-time data scientist.

Key risks: vandalism/theft, false positives, governance delays. Mitigations: community guardians, human-in-the-loop, pre-signed templates.

50k USD — Medium (Full Corridor Pilot)

Full corridor implementation with stronger coverage and safeguards. Deploy 8 count-only camera sites, 50 sensor kits, and 10 rain gauges, with a scalable pipeline. Add selective VHR/drone validation and link alerts to predefined response actions (mediation coordination, water repair, targeted vet support). Maintain strict privacy: aggregated outputs only, no exact water points, role-based access, and consent-first operations.

Timeline & staffing: 9 months; staffed project lead, local ops manager, 2 full-time field techs, data scientist, protection officer.

Key risks: political sensitivity and data misuse. Mitigations: strict DSAs, anonymization, emergency extraction fund.

STABILITY & LIVELIHOODS

We protect livelihoods before tensions escalate.

Pastoral systems rely on mobility. When herd movements become compressed or unpredictable, resource competition intensifies and social friction rises.

By identifying emerging convergence risk early, our system supports proactive coordination rather than reactive enforcement. Protecting livestock access protects household income, food security, and inter-community trust.

Reducing uncertainty reduces conflict pressure.



On a more personal note...

Dear United Nations Team,

We are grateful for the opportunity to propose a solution to a challenge that impacts safety, livelihoods, and stability in pastoral regions. Our project focuses on how water scarcity and changing conditions can influence herd movement and increase the risk of local conflict, and how open-source satellite and community signals can support earlier coordination and response. Through this challenge, each of us has grown significantly. From learning the realities of pastoral systems, exploring the limits of satellite data, speaking with practitioners, and pressure testing our assumptions, this work has been both challenging and meaningful. If you have any questions about our proposal or would like to discuss the pilot design, safeguards, or scalability, please feel free to reach out.

Thank you,
Abubakr, Abyad, Dayem, Nia



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Appendix

[Github Repository](#)

[Methodology
Documentation](#)



[Researches &
Studies](#)